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School of Engineering

Advanced Composite Materials 5
Design Review

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Contents

1 Part I: Analysis of a composite plate subjected to uniaxial loading **1**
1.1 Composite Laminate Plate Theory Calculations . . . 1
1.1.1 Deriving the ABD Matrices . . . 1
1.1.2 Obtaining laminate strains . . . 2
1.1.3 Obtaining ply strains and ply stresses . . . 2
1.2 Comparison to eLamX^2 results . . . 3
1.2.1 Laminate-level stiffness and strains . . . 4
1.2.2 Ply-level strains and stresses . . . 4
1.2.3 Fibre failure . . . 5

2 Layup design for composite plate with a hole under tension and shear **6**
2.1 Safety Factor using Tsai-Wu failure criteria . . . 6
2.2 Safety improvements using Composite Laminate Plate Theory . . . 7
2.3 Limitations of FEA model . . . 9
2.4 Other laminate improvement methods . . . 9

References **11**

List of Figures

1 eLamX^2 force analysis of quasi-isotropic plate including ABD matrix . . . 4
2 eLamX^2 layer information in the local unidirectional coordinate system - Strains . . . 4
3 eLamX^2 layer information in the local unidirectional coordinate system - Stresses . . . 5
4 eLamX^2 stress-strain distributions . . . 5
5 Plate test scenarios . . . 6
6 Abaqus results from shell load tension test . . . 6
7 Abaqus results from concentrated shear being test . . . 7
8 Abaqus results of laminate layup B . . . 8
9 Test specimen with metal inserts and a schematic sketch showing the layup of inserted metal patches . . . 10

List of Tables

1 Elastic Properties of the CFRP Laminate Plate . . . 1
2 Ply strains . . . 3
3 Ply Stresses . . . 3
4 Strength Properties of the CFRP laminate plate . . . 4
5 Comparison of Tsai-Wu safety factors for baseline and redesigned layups . . . 8

Notation

E_1	–	Longitudinal elastic modulus
E_2	–	Transverse elastic modulus
G_{12}	–	In-plane shear modulus
G_{13}	–	Shear modulus in the 1–3 plane (longitudinal–through-thickness)
G_{23}	–	Shear modulus in the 2–3 plane (transverse–through-thickness)
ν_{12}	–	Major Poisson's ratio
ν_{12}	–	Major Poisson's ratio (transverse strain due to a longitudinal stress)
ν_{21}	–	Minor Poisson's ratio (longitudinal strain due to a transverse stress)
Q_{ij}	–	Reduced stiffness matrix components (lamina coordinates)
$\bar{Q}_{ij}$	–	Transformed reduced stiffness matrix components
A	–	Extensional stiffness matrix
B	–	Bending–extension coupling matrix
D	–	Bending stiffness matrix
z_k	–	Through-thickness coordinate of kth ply
N_x, N_y, N_{xy}	–	In-plane load resultants
M_x, M_y, M_{xy}	–	Moment resultants
$\varepsilon_x^0, \varepsilon_y^0, \gamma_{xy}^0$	–	Mid-plane engineering strains
$\kappa_x, \kappa_y, \kappa_{xy}$	–	Curvatures
$\varepsilon_1, \varepsilon_2, \gamma_{12}$	–	Ply strains in local 1–2 axes
σ_1	–	Fibre-direction stress
σ_2	–	Transverse stress
τ_{12}	–	In-plane shear stress
σ_1^T, σ_1^C	–	Longitudinal tensile/compressive strength
σ_2^T, σ_2^C	–	Transverse tensile/compressive strength
F_{12}	–	Tsai-Wu interaction coefficient
TW	–	Tsai-Wu failure index
SF	–	Safety factor
t	–	Ply thickness
θ	–	Ply orientation angle

1 Part I: Analysis of a composite plate subjected to uniaxial loading

1.1 Composite Laminate Plate Theory Calculations

1.1.1 Deriving the ABD Matrices

A quasi-isotropic composite plate $[0/\pm45/90]_s$ with a ply thickness of t = 0.2 mm experiences a shell edge load of 300 N/mm. The composite plate, therefore, comprises 8 plies in the layup 0, +45, -45, 90, 90, -45, -45 ,0, giving a total thickness of 1.6 mm. Where the midplane is z = 0, the top surface is at z = -0.8 mm and top surface is at z = 0.8 mm. Therefore, the ply interface coordinates can be expressed by Equation 1. The elastic material properties are given in Table 1.

$$z_k = 0.2k - 0.8 \quad (k = 0, ..., 8) \tag{1}$$

Table 1: Elastic Properties of the CFRP Laminate Plate

E_1	E_2	G_{12}	G_{13}	G_{23}	ν_{12}
140 GPa	7 GPa	4 GPa	4 GPa	4 GPa	0.3

In the lamina 1-2 coordinate system, the reduced stiffness matrix $[Q]$ is:

$$[Q] = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix}$$

Using the propeties given by Table 1, values Q_{11}, Q_{12}, Q_{22}, and Q_{66} are calculated as follows:

$$Q_{11} = \frac{E_1}{1 - \nu_{12}\,\nu_{21}} = \frac{E_1}{1 - \nu_{12}^2\left(\frac{E_2}{E_1}\right)} = \frac{140}{1 - 0.3^2 \times \left(\frac{7}{140}\right)} = 140.63 \text{ GPa} \tag{2}$$

$$Q_{12} = \frac{E_2}{1 - \nu_{12}\,\nu_{21}} = \frac{E_2}{1 - \nu_{12}^2\left(\frac{E_2}{E_1}\right)} = \frac{7}{1 - 0.3^2 \times \left(\frac{7}{140}\right)} = 7.03 \text{ GPa} \tag{3}$$

$$Q_{22} = \frac{\nu_{12}\,E_1}{1 - \nu_{12}\,\nu_{21}} = \frac{\nu_{12}\,E_2}{1 - \nu_{12}^2\left(\frac{E_2}{E_1}\right)} = \frac{0.3 \times 7}{1 - 0.3^2 \times \left(\frac{7}{140}\right)} = 2.11 \text{ GPa} \tag{4}$$

$$Q_{66} = G_{12} = 4 \text{ GPa} \tag{5}$$

$$\therefore \quad [Q] = \begin{bmatrix} 140.63 & 7.03 & 0 \\ 7.03 & 2.11 & 0 \\ 0 & 0 & 4.00 \end{bmatrix} \text{ GPa}$$

For a ply oriented at an angle θ to the global x-axis, the transformed reduced stiffness matrix $[\bar{Q}]$ is constructed using the following laminate theory relations, where $m = \cos\theta$ and $n = \sin\theta$:

$$\bar{Q}_{11} = m^4 Q_{11} + 2m^2n^2(Q_{12} + 2Q_{66}) + n^4 Q_{22},$$

$$\bar{Q}_{12} = m^2n^2(Q_{11} + Q_{22} - 4Q_{66}) + Q_{12}(m^4 + n^4),$$

$$\bar{Q}_{22} = n^4 Q_{11} + 2m^2n^2(Q_{12} + 2Q_{66}) + m^4 Q_{22},$$

$$\bar{Q}_{16} = m^3n(Q_{11} - Q_{12}) + mn^3(Q_{12} - Q_{22}) - 2mn(m^2 - n^2)Q_{66},$$

$$\bar{Q}_{26} = mn^3(Q_{11} - Q_{12}) + m^3n(Q_{12} - Q_{22}) + 2mn(m^2 - n^2)Q_{66},$$

$$\bar{Q}_{66} = m^2n^2(Q_{11} + Q_{22} - 2Q_{12}) + (m^2 - n^2)^2 Q_{66}.$$

The reduced stiffness matrices for the four ply orientation in the laminate are:

$$[\bar{Q}]_{0^\circ} = \begin{bmatrix} 140.63 & 2.11 & 0 \\ 2.11 & 7.03 & 0 \\ 0 & 0 & 4 \end{bmatrix} \text{ GPa} \qquad [\bar{Q}]_{+45^\circ} = \begin{bmatrix} 41.97 & 33.97 & 33.40 \\ 33.97 & 41.97 & 33.40 \\ 33.40 & 33.40 & 35.86 \end{bmatrix} \text{ GPa}$$

$$[\bar{Q}]_{-45^\circ} = \begin{bmatrix} 41.97 & 33.97 & -33.40 \\ 33.97 & 41.97 & -33.40 \\ -33.40 & -33.40 & 35.86 \end{bmatrix} \text{ GPa} \qquad [\bar{Q}]_{90^\circ} = \begin{bmatrix} 7.03 & 2.11 & 0 \\ 2.11 & 140.63 & 0 \\ 0 & 0 & 4 \end{bmatrix} \text{ GPa}$$

The ABD matrices are given by the following definitions:

$$[A] = \sum_{k=1}^{N} \left[\bar{Q}\right]^{(k)} (z_k - z_{k-1}) \qquad [B] = \frac{1}{2} \sum_{k=1}^{N} \left[\bar{Q}\right]^{(k)} (z_k^2 - z_{k-1}^2) \qquad [D] = \frac{1}{3} \sum_{k=1}^{N} \left[\bar{Q}\right]^{(k)} (z_k^3 - z_{k-1}^3)$$

Because the laminate is symmetric, [B] = 0, and bending-stretching coupling disappears. Matrices [A] and [B] are calculated to give the following:

$$[A] = \begin{bmatrix} 92.6 & 2.89 & 0 \\ 28.9 & 92.6 & 0 \\ 0 & 0 & 31.9 \end{bmatrix} \text{ kN/mm (GPa)} \qquad [D] = \begin{bmatrix} 33.6 & 5.14 & 2.14 \\ 5.14 & 7.96 & 2.14 \\ 2.14 & 2.14 & 5.78 \end{bmatrix} \text{ kN/mm (GPa)}$$

1.1.2 Obtaining laminate strains

Assuming that the shell load is along the x-axis, and there is no load in the y-axis or in-plane shear, the in-plane shell edge load is N_x = 0.3 kN/mm for unit consistency with the [A] and [B] matrices. A shell edge load is simply a load per unit length. This can be written as:

$$\{N\} = \begin{bmatrix} 0.3 \\ 0 \\ 0 \end{bmatrix} \text{ kN/mm}$$

The relation governing the in-plane behaviour of the laminate is

$$\{N\} = [A]\{\varepsilon^0\} + [B]\{\kappa\} \tag{6}$$

As [B] = 0 and there is no bending moments (M = 0):

$$\{M\} = [B]\{\varepsilon^0\} + [D]\{\kappa\} \Rightarrow \{\kappa\} = 0 \tag{7}$$

Using the value for [A] calculated in Section 1.1.1, the strains are calculated as follows:

$$\{\varepsilon^0\} = \begin{bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{bmatrix} = [A]^{-1}\{N\} = \begin{bmatrix} 1.247 \times 10^{-2} & -3.893 \times 10^{-3} & 0 \\ -3.893 \times 10^{-3} & 1.247 \times 10^{-2} & 0 \\ 0 & 0 & 3.134 \times 10^{-2} \end{bmatrix} \begin{bmatrix} 0.3 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3.59 \times 10^{-3} \\ -1.12 \times 10^{-3} \\ 0 \end{bmatrix} \tag{8}$$

Therefore, the laminate strain values are ε_x^0 = 3.59 $\times$ 10^{-3}, ε_y^0 = -1.12 $\times$ 10^{-3}, and γ_{xy}^0 = 0.

1.1.3 Obtaining ply strains and ply stresses

The global mid-plane engineering strain vector was derived in Equation 8. Because the laminate is symmetric and subjected only to in-plane loading, the curvatures are zero and these global strains are uniform throughout the thickness. Therefore, the in-plane global strains in each ply are equal to the mid-plane values.

For a ply oriented at an angle θ to the global x-axis, the local (1-2) strain tensor is obtained by tensor transformation. The global strain tensor is written as:

$$[\varepsilon]_{xy} = \begin{bmatrix} \varepsilon_x^0 & \gamma_{xy}^0/2 \\ \gamma_{xy}^0/2 & \varepsilon_y^0 \end{bmatrix} \tag{9}$$

Using the rotation matrix relating the x-y axis to the local 1-2 axis

$$[R] = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \tag{10}$$

the local strain tensor is then $[\varepsilon] = R^T[\varepsilon]_{xy}R$. From this tensor the local engineering strains are extracted as $\varepsilon_1 = \varepsilon_{11}$, $\varepsilon_2 = \varepsilon_{22}$, and $\gamma_{12} = 2\varepsilon_{12}$. Applying this procedure gives the ply strains shown in Table 2.

Table 2: Ply strains

	ε_1	ε_2	γ_{12}
0°	3.59 $\times 10^{-3}$	-1.12 $\times 10^{-3}$	0
45°	1.23 $\times 10^{-3}$	1.23 $\times 10^{-3}$	-4.70 $\times 10^{-3}$
-45°	1.23 $\times 10^{-3}$	1.23 $\times 10^{-3}$	4.70 $\times 10^{-3}$
90°	-1.12 $\times 10^{-3}$	3.59 $\times 10^{-3}$	0

The ply stresses can be calculated using Equation 11 and the unidirectional lamina stiffness matrix $[Q]$ as derived in Section 1.1.1. The results are shown in Table 3.

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = [Q] \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{bmatrix} \tag{11}$$

Table 3: Ply Stresses

	σ_1 [MPa]	σ_2 [MPa]	τ_{12} [MPa]
0°	5.02 $\times 10^2$	-0.29	0
+45°	1.76 $\times 10^2$	11.3	18.8
-45°	1.76 $\times 10^2$	11.3	-18.8
90°	-1.50 $\times 10^2$	22.9	0

These results show that the laminate response under the applied uniaxial load is dominated by the 0° plies, which carry the majority of the axial stress through the fibres, while in-plane shear occurs in the $\pm$45° plies. The 90° plies contribute little to the axial stiffness.

This distribution is characteristic of a balanced and symmetric lay-up and allows direct evaluation of likely failure modes in each orientation. These CLPT results provide a validated analytical foundation for the laminate-level and ply-level failure assessment carried out in Section 1.2 and for comparison against eLamX^2.

1.2 Comparison to eLamX^2 results

The CLPT results derived in Section 1.1 can be compared to results produced by eLamX^2, an open-source composite calculator that bases its calculations on the same classical laminated plate theory [1].

The same quasi-isotropic composite plate [0/$\pm$45/90]$_s$ is set up using the same E_1, E_2, G_{12}, and ν_{12} values given in Table 1, as well as the strength properties in Table 4. All other values remained as default, which may induce errors in other analyses other than those discussed in this report. A new laminate is created with this material, in the layup specified above, with a ply thickness of 0.2 mm. The shell edge load N_x = 300 N/mm was applied.

Table 4: Strength Properties of the CFRP laminate plate

σ_1^T	σ_1^C	σ_2^T	σ_2^C	τ_{12}	F_{12} Coefficient (Tsai-Wu failure criteria)
2500 MPa	1200 MPa	80 MPa	210 MPa	120 MPa	-0.5

1.2.1 Laminate-level stiffness and strains

Figure 1 shows the ABD matrix and strain vector computed by eLamX^2. Comparing the manually calculated values from Section 1.1.1, it can be seen that the results are nearly identical, with the largest being 0.04 GPa (0.14%) in A_{12}, all others differentiated by less than 0.1%.

These differences can be attributed to rounding in the manual calculations leading up to the derivation of the ABD matrix, primarily the Q matrix which was expressed in GPa to 3 significant figures, rather than the SI standard of N/mm (MPa). If it had been written in MPa to 2 decimal places the ABD matrix and other derived values would be more accurate. When doing the same calculations, eLamX^2 will store values to much higher accuracy.

The ε_x, ε_y, and γ_{xy} strains similarly calculated by eLamx^2 are near identical to the manual calculations. The differences again are due to rounding rather than modelling errors.

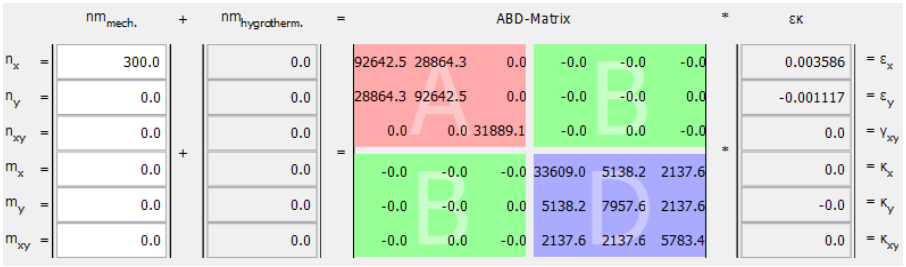


Figure 1: eLamX^2 force analysis of quasi-isotropic plate including ABD matrix

1.2.2 Ply-level strains and stresses

Figures 2 and 3 show the calculated layer information in the local 1-2 unidirectional coordinate system. These can be compared directly with the manually derived ply strains (Table 2) and ply stresses (Table 3). For each orientation, the numerical values of ε_1, ε_2, γ_{12} and σ_1, σ_2 and τ_{12} match within rounding. For example the 0° plies carry an axial stress of about 502 MPa, while the $\pm$45° plies experience 176 MPa in the fibre direction and $\pm$18.8 MPa in shear in both the manual calculations and the eLamX^2 output.

No.	Angle	z_i	$\varepsilon_{\|\|}$	$\varepsilon_{\perp}$	$\gamma_{\|\|\perp}$	Failure Criteria	RF_{min}	Failure Type
1	0.0	0.8	0.003586	-0.001117	0.0	TsaiWu (Abaqus)	2.983	Failure
		0.6	0.003586	-0.001117	0.0	TsaiWu (Abaqus)	2.983	Failure
2	45.0	0.6	0.001234	0.001234	-0.004704	TsaiWu (Abaqus)	6.351	Failure
		0.4	0.001234	0.001234	-0.004704	TsaiWu (Abaqus)	6.351	Failure
3	-45.0	0.4	0.001234	0.001234	0.004704	TsaiWu (Abaqus)	6.351	Failure
		0.2	0.001234	0.001234	0.004704	TsaiWu (Abaqus)	6.351	Failure
4	90.0	0.2	-0.001117	0.003586	-0.0	TsaiWu (Abaqus)	4.065	Failure
		-0.0	-0.001117	0.003586	-0.0	TsaiWu (Abaqus)	4.065	Failure
5	90.0	-0.0	-0.001117	0.003586	-0.0	TsaiWu (Abaqus)	4.065	Failure
		-0.2	-0.001117	0.003586	-0.0	TsaiWu (Abaqus)	4.065	Failure
6	-45.0	-0.2	0.001234	0.001234	0.004704	TsaiWu (Abaqus)	6.351	Failure
		-0.4	0.001234	0.001234	0.004704	TsaiWu (Abaqus)	6.351	Failure
7	45.0	-0.4	0.001234	0.001234	-0.004704	TsaiWu (Abaqus)	6.351	Failure
		-0.6	0.001234	0.001234	-0.004704	TsaiWu (Abaqus)	6.351	Failure
8	0.0	-0.6	0.003586	-0.001117	0.0	TsaiWu (Abaqus)	2.983	Failure
		-0.8	0.003586	-0.001117	0.0	TsaiWu (Abaqus)	2.983	Failure

Figure 2: eLamX^2 layer information in the local unidirectional coordinate system - Strains

No.	Angle	z_i	$\sigma_{\|\|}$	$\sigma_{\perp}$	$\tau_{\|\|\perp}$	Failure Criteria	RF_{min}	Failure Type
1	0.0	0.8	502.008	-0.292	0.0	TsaiWu (Abaqus)	2.983	Failure
		0.6	502.008	-0.292	0.0	TsaiWu (Abaqus)	2.983	Failure
2	45.0	0.6	176.215	11.285	-18.815	TsaiWu (Abaqus)	6.351	Failure
		0.4	176.215	11.285	-18.815	TsaiWu (Abaqus)	6.351	Failure
3	-45.0	0.4	176.215	11.285	18.815	TsaiWu (Abaqus)	6.351	Failure
		0.2	176.215	11.285	18.815	TsaiWu (Abaqus)	6.351	Failure
4	90.0	0.2	-149.578	22.861	-0.0	TsaiWu (Abaqus)	4.065	Failure
		-0.0	-149.578	22.861	-0.0	TsaiWu (Abaqus)	4.065	Failure
5	90.0	-0.0	-149.578	22.861	-0.0	TsaiWu (Abaqus)	4.065	Failure
		-0.2	-149.578	22.861	-0.0	TsaiWu (Abaqus)	4.065	Failure
6	-45.0	-0.2	176.215	11.285	18.815	TsaiWu (Abaqus)	6.351	Failure
		-0.4	176.215	11.285	18.815	TsaiWu (Abaqus)	6.351	Failure
7	45.0	-0.4	176.215	11.285	-18.815	TsaiWu (Abaqus)	6.351	Failure
		-0.6	176.215	11.285	-18.815	TsaiWu (Abaqus)	6.351	Failure
8	0.0	-0.6	502.008	-0.292	0.0	TsaiWu (Abaqus)	2.983	Failure
		-0.8	502.008	-0.292	0.0	TsaiWu (Abaqus)	2.983	Failure

Figure 3: eLamX^2 layer information in the local unidirectional coordinate system - Stresses

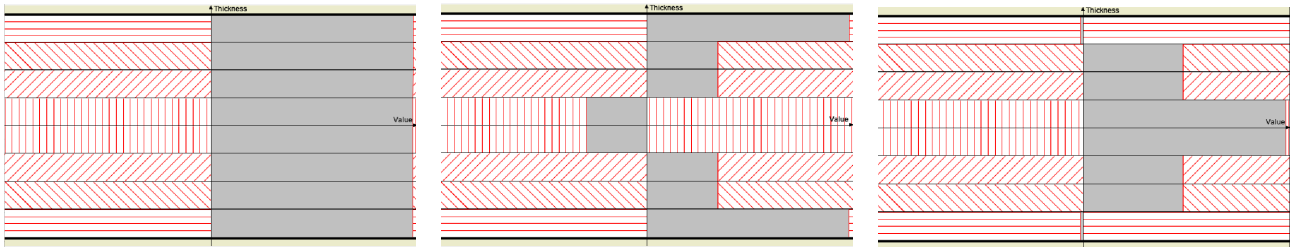
The comparison verifies the load-sharing behaviour of a quasi-isotropic lay-up under uniaxial tension: the 0° plies dominate the axial stiffness and stress, the $\pm$45° plies accommodate in-plane shear, and the 90° plies primarily experience transverse tension/compression with relatively low axial stress. This pattern is consistently reproduced by both approaches.

1.2.3 Fibre failure

eLamX^2 was also used to evaluate the Tsai-Wu failure indices for each ply. The innermost 90° plies were identified as the most critical, with the lowest reserve factor (safety factor) of 2.69. This is a Tsai-Wu failure value of 0.386. Failure occurs at a Tsai-Wu value of 1, so no fibres fail, but the inner 90° plies would fail first under an increased shell load, by mode of matrix cracking and transverse tension–compression failure with possible delamination at the interface of the $\pm$45° plies.

Once the 90° plies crack, their transverse stiffness drops, so more of the load is taken up by the remaining plies. The next failure would be in the 0°, which has the next lowest safety factor of 4.972, and undergoes fibre-dominated tensile failure. At this point the axial stiffness drops and the laminate has effectively failed for most design purposes. The $\pm$45° plies fail last with a safety factor of 5.29, by method of matrix shear cracking and fibre-shear failure.

Figure 4 shows how the stress and strain develop through the thickness of the quasi-isotropic laminate under the applied shell load. The global x-strain distribution in Figure 4a shows that ε_x^0 remains constant throughout the thickness, in theory. Figure 4b shows that the 0° plies carry the highest axial stress, the $\pm$45° plies a lower amount, and the 90° plies experience a small amount of compressive load, due to the Poisson effect. Figure 4c shows the transverse stress σ_2, which is the largest in the 90° plies, explaining how the Tsai-Wu failure index identifies these plies as the first to fail under increasing load.

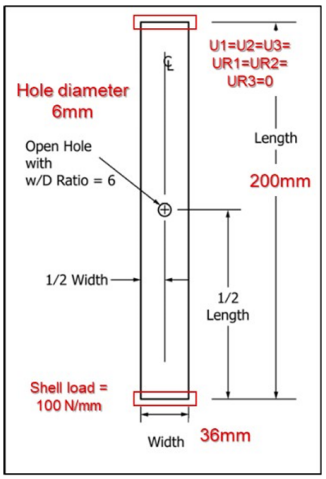


(a) Global x-component strain ε_x^0 (b) Local fibre-direction stress σ_1 (c) Local transverse stress σ_2

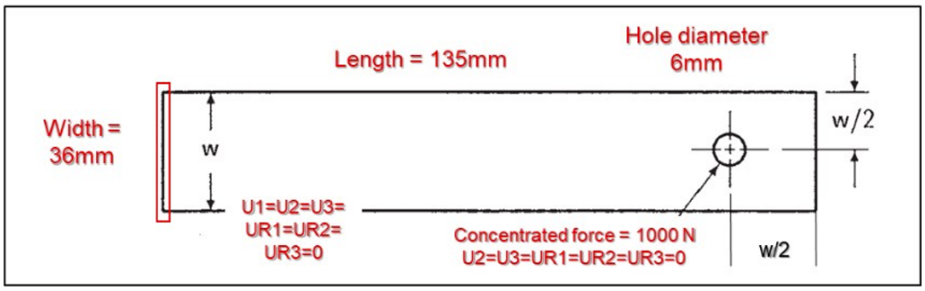
Figure 4: eLamX^2 stress-strain distributions

2 Layup design for composite plate with a hole under tension and shear

In Abaqus, finite element analysis (FEA) is performed on a composite plate with a layup $[(0/90/90/90)_2]_s$ that is 16 plies thick [2]. Each unidirectional ply is 0.15 mm thick, giving a total thickness of 2.4 mm. As shown in Figure 5, this layup will be simulated under a shell load of 100 N/mm and a concentrated 1000 N to induce a shearing load. The laminate plates were constructed in Abaqus using the elastic properties given in Table 1 and 4, assuming the fibre direction aligns with the length dimension.



(a) Open hole tension, shell load



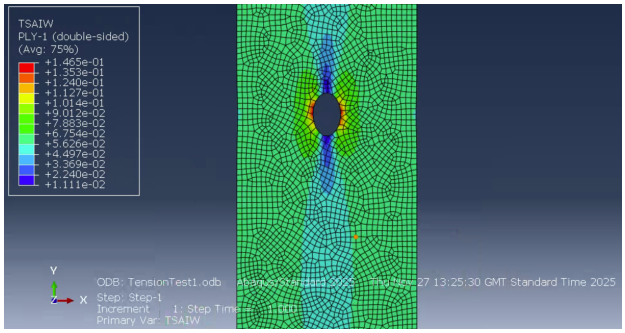
(b) Shear bearing, concentrated load

Figure 5: Plate test scenarios

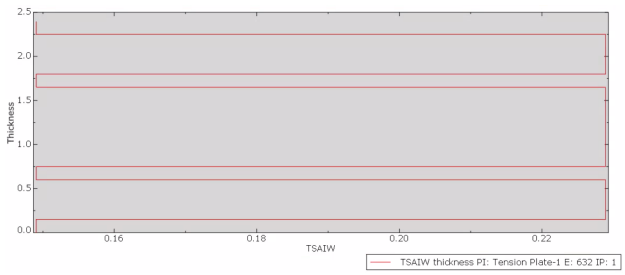
2.1 Safety Factor using Tsai-Wu failure criteria

The results of the tension set-up shown in Figure 5a show that the safety factor, equal to the inverse of the highest Tsai-Wu failure criteria value 0.2258, which occured in the 90° ply, is 4.43. The 0° ply exhibited a Tsai-Wu failure criteria value of 0.1465, which is a 6.83 safety factor in this ply, 54% higher than the 90° ply. Figure 6a shows the Tsai-Wu failure criteria on the outermost ply. It can be seen that there are concentrations of stress both side of the hole, with less stress directly above and below the hole due to the Poisson effect. Figure 6b shows the Tsai-Wu distribution throughout the laminate thickness, taken at an element to the side of the hole.

As all Tsai–Wu indices remain below 1.0, no failure is predicted under the applied 100 N/mm shell load. The critical 90° ply is governed by transverse matrix tension/compression around the hole, with local σ_2 and τ_{12} driving the Tsai–Wu index rather than fibre-direction stress.



(a) Tsai-Wu failure criteria on 1st (0°) ply



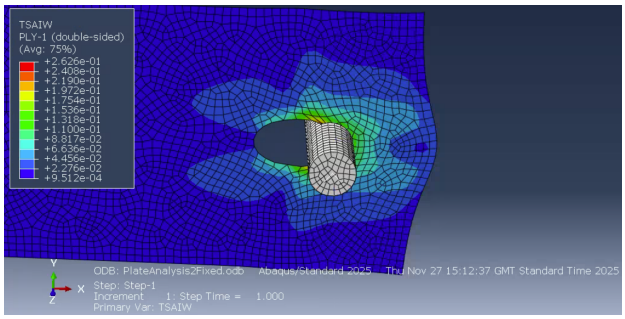
(b) Tsai-Wu failure criteria through laminate thickness

Figure 6: Abaqus results from shell load tension test

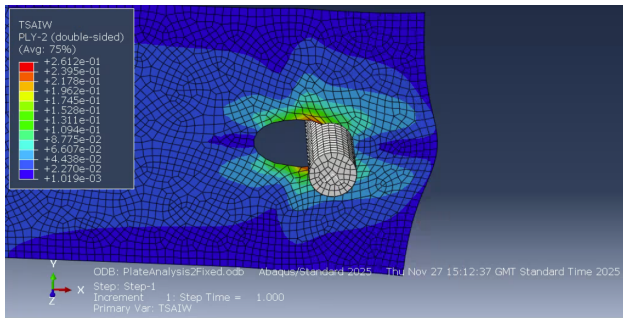
The results of the shear bearing set-up shown in Figure 5b show that the safety factor, inverse of the highest Tsai-Wu failure criteria value 0.2626, in the 0° ply, is 3.81, lower than in the tension test by 14%,

although the specimen lengths and hole location differ. The 90° ply exhibited a Tsai-Wu failure criteria value of 0.2612, which is a 3.83 safety factor in this ply, just 0.5% higher than the 0° ply. In Figure 7a and 7b, which show the Tsai-Wu failure values across the outermost 0° ply and 90° ply, it can be seen for the 0° ply, the highest level of stress occurs on the right-hand side of the pin, while for the 90° ply, the highest stress occurs in two separate concentrations above and below the pin. Figure 7c shows the Tsai-Wu failure criteria through the thickness. The element chosen was the one with the highest Tsai-Wu failure criteria in the 0° ply, so the values recorded for the 90° ply may have some inaccuracy. This also applies to Figure 7d, which shows the stresses through the laminate thickness.

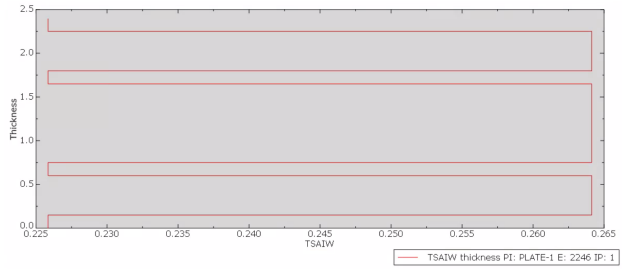
Again, all plies have a Tsai–Wu index below 1.0, so no failure is predicted for the 1000 N shear bearing load. The near-identical critical indices in the 0° and 90° plies indicate a mixed failure tendency: fibre-direction shear in the 0° plies and matrix-dominated shear/tension in the 90° plies around the pin–hole contact region.



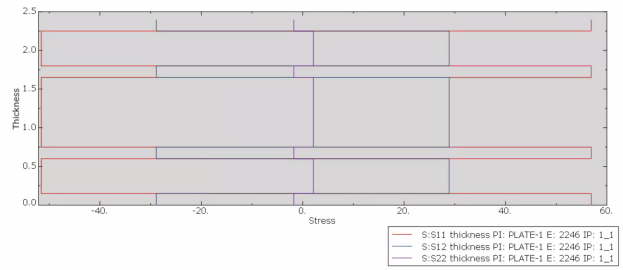
(a) Tsai-Wu failure criteria on 1st (0°) ply



(b) Tsai-Wu failure criteria on 2nd (90°) ply



(c) Tsai-Wu failure criteria through laminate thickness



(d) S_{12}, S_{12}, S_{22} stresses through laminate thickness

Figure 7: Abaqus results from concentrated shear being test

2.2 Safety improvements using Composite Laminate Plate Theory

With a base safety factor of 4.43 in the tension simulation and 3.81 in the shear bearing simulation, any changes in the laminate layup should increase the safety factor in both scenarios.

In the tension simulation, the limitation was the 90° ply, providing little resistance to longitudinal load. The 0° has a safety factor 54% higher. With twelve 90° plies and just four 0° plies, the original laminate is very weak in tension, which is dominated by the σ_1 stiffness. In the shear simulation, the 0° and 90° had nearly equal safety factors, indicating shear-dominated matrix failure.

Based on the weaknesses of the original layup, a new layup is required that:

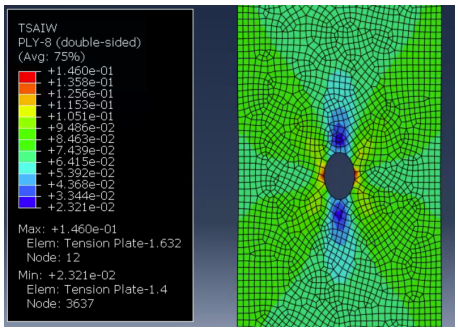
1. Reduces σ_2 and τ_{12} in the 90° plies
2. Introduces more $\pm 45°$ plies for better shear load carrying
3. Introduces more 0° plies for tension strength

Based on these requirements and the restriction that the number of plies must remain at 16 and the thickness of each be 0.15 mm, five possible layups were tested in Abaqus. These are listed in Table 5.

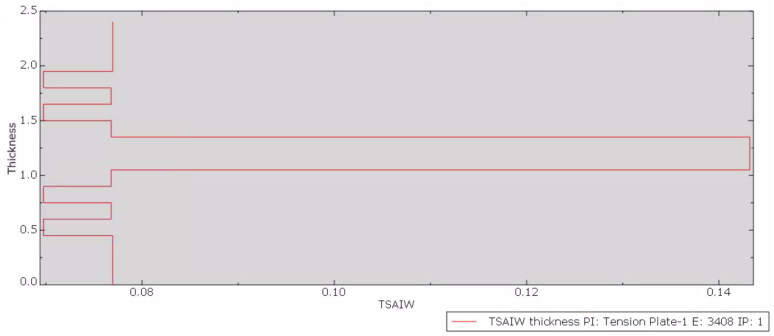
Table 5: Comparison of Tsai-Wu safety factors for baseline and redesigned layups

Layup	Name	Tension			Shear		
		Tsai-Wu	SF	Weakest Ply	Tsai-Wu	SF	Weakest Ply
$[(0/90/90/90)_2]_s$	Original	0.2258	4.43	90°	0.2626	3.81	0°
$[(0/\pm45/90)_2]_s$	A	0.1733	5.77	90°	0.1902	5.26	0°
$[(0/0/0/\pm45/\pm45/90)]_s$	B	0.1460	6.84	90°	0.1816	5.51	0°
$[(0/\pm45/\pm45/\pm45/90)]_s$	C	0.2657	3.76	90°	0.2613	3.83	0°
$[(0/0/0/0/\pm45/\pm45)]_s$	D	0.06978	14.33	0°	0.2111	4.74	0°
$[(0/\pm45/\pm45/\pm45/0)]_s$	E	0.07988	12.52	±45°	0.2582	3.87	0°

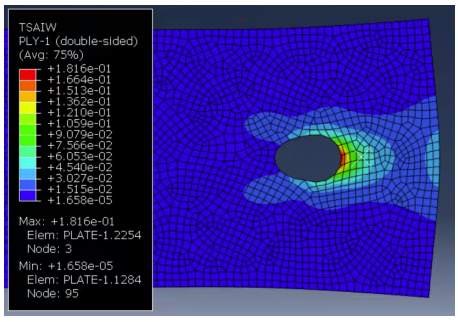
Layup A $[(0/\pm45/90)_2]_s$ is a quasi-isotropic layup that theoretically should give equal properties in each direction. Layup B $[(0/0/0/\pm45/\pm45/90)]_s$ is similar but provides higher strength in the longitudinal direction, taking strength away from the transverse direction. Both layups reduce the maximum Tsai-Wu index in both cases, meaning the additional 0° are transferring stress from the 90° plies. The weakest plies remain the same in each test scenario, but the safety factors have both been increased. In shear, the higher safety factor shows that the ±45° plies have increased the laminate shear stiffness A_{66} in the ABD matrix, so less τ_{12} is carried in the 0° plies. Layup B is the best all-round design, with both safety factors increased with over-designing one load case at the expense of the other. Results are shown in Figure 8.



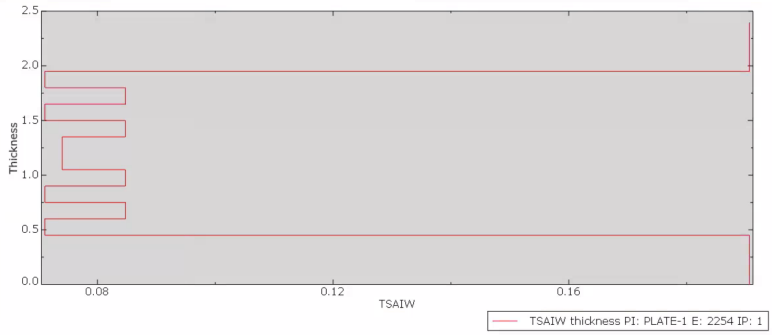
(a) FEA results of 90° ply, tension load case



(b) Tsai-Wu failure criteria through thickness, tension load case



(c) FEA results of 90° ply, shear load case



(d) Tsai-Wu failure criteria through thickness, shear load case

Figure 8: Abaqus results of laminate layup B

Layup C $[(0/\pm45/\pm45/\pm45/90)]_s$ adds in more ±45° plies, 12 total, with an equal number of 0° and 90° plies. In tension, the safety factor drops to 3.76, making it worse than the original, while the shear safety factor is almost unchanged at 3.83. This makes it an unsuitable layup, and shows that just adding ±45° plies is not enough as σ_2 around the hole can still drive Tsai-Wu failure.

Layup D $[(0/0/0/0/\pm45/\pm45)]_s$ and E $[(0/\pm45/\pm45/\pm45/0)]_s$ are both tension focused with no 90° plies in either layup. Tension safety factors jump to 14.33 and 12.52, and the weakest plies switch from 90° for layup D and ±45° for layup E, as the 90° plies cannot be the limited layers, as there aren't any in the laminate. However, the shear safety factors are only 4.74 for D and 3.87 for E, thus D is better than E for both

load cases, but worse than A and B, while E is only marginally better than the original. So these layups are over-designed for tension, and would be inefficient for a design where both load cases are important.

The results show that replacing the 90° ply content reduces the matrix-dominated failure in tension, and adding ±45° plies improves shear performance by distributing τ_{12}, but only if enough 0° plies remain to keep A_{11} high. Layup B provides the best compromise of safety factors, 6.84 in tension and 5.51 in shear, with a 54.4% and 44.6% increase without extreme tension over-capacity. In the case where tensile loading is in the non-lengthways direction, a quasi-isotropic layup like A may be better, as layup B is optimised for tensile load in the lengthways direction only.

2.3 Limitations of FEA model

The finite element analysis used in this study provides the useful ability to compare the characteristics of different laminate layups. However, there are numerous important limitations, as it is an idealistic model.

1. Perfect bonding between plies is assumed. Abaqus models the laminate as perfectly bonded with no possibility of interface failure or delamination. In physical specimens, delamination around holes is common due to interlaminar shear and transverse tension. Because the model cannot simulate delamination propagation, the laminate strength is overestimated.

2. Manufacturing defects. The model assumes defect-free plies with perfect alignment. Real laminates contain fibre waviness, misalignment, porosity, resin-rich pockets, and residual stresses from curing. These defects reduce transverse strength σ_2, shear strength τ_{12}, and interlaminar properties, the modes of failure that controlled the Tsai-Wu index in the original laminate layup. Therefore, physical specimens would fail before the model prediction.

3. Damage from hole drilling. Mechanically drilled holes will introduce micro-cracking, interface debondings, and frayed fibres at the bore surface. These create local stresses and weaken the laminate where the Tsai-Wu index is highest. This is a significant omission of the model, as cracks that propagate can cause failure at much lower loads than predicted.

4. Limitations of shell elements. The shell elements in Abaqus assume a linear through-thickness stress field and cannot capture interlaminar normal stress σ_{zz} or interlaminar shear τ_{xz} and τ_{yz}. A 3D continuum model would give more realistic failure predictions, but with a higher computation cost.

5. No progressive damage modelling. Tsai-Wu is applied instantaneously and does not simulate stiffness degradation, creep, fibre kinking, load redistribution, or delamination growth. In a physical specimen, once a 90° ply begins to crack or a ±45° ply shears, the local stiffness will drop, and damage accelerates. The FEA predicts the first ply failure only, not the full damage progression.

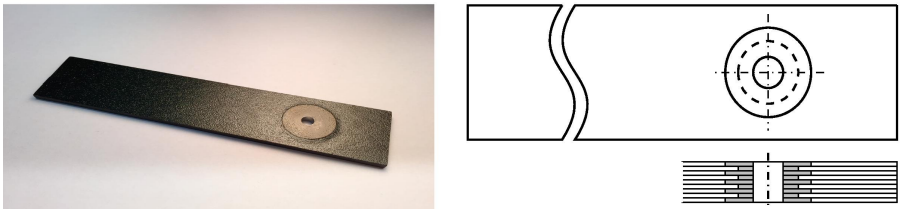
6. Mesh sensitivity. Stress concentrations around holes are highly mesh sensitive. If the mesh is not sufficiently refined locally, peak stresses may be underpredicted.

In summary, these limitations mean that the FEA model overpredicts laminate strength and safety factors. While it is useful for comparing different layups, physical testing should always be conducted to validate the performance of any laminate design.

2.4 Other laminate improvement methods

There are several methods of improving the safety factor of the laminate and changing the failure mode. These include local reinforcements, hybrid material systems, alternative fibre architectures, and advanced manufacturing technologies.

Peak stress occurred around the hole, an order of magnitude higher than the rest of the plate, so reducing the stress around the hole is critical. This can be done with local hole reinforcement with high-strength metal patches between selected plies, which have been found to give up to a 60% strength improvement in pin-loaded tests where the 0° plies meet metal patches with overlapping interfaces [3]. Joints become marginally stronger if smaller patches are used in the outermost plies rather than larger ones. Metal inserts directly carry bearing loads and redistribute stresses away from weaker matrix-dominated plies. Figure 9 shows a test specimen with metal inserts and a schematic illustrating the internal layup [3].



(a) (b)

Figure 9: (a) Test specimen with metal inserts (b) schematic sketch showing the layup of inserted metal patches

Hybrid laminates, such as CFRP–titanium systems containing around 20% titanium by volume, have demonstrated 32% higher specific tensile strength in multi-row bolted joints compared with pure CFRP [4]. Hybrid effects also increase failure strain for very thin plies, up to 20% [5]. This occurs because the metallic component provides higher ductility and local plastic deformation, delaying crack initiation and stabilising damage propagation. Selectively hybridising outer plies or those adjacent to hole surfaces can therefore enhance both strength and toughness without significantly altering global stiffness.

Shear failure and delamination can also be mitigated by through-thickness reinforcements introduced during manufacturing, such as simultaneous stitching, which simply stitches the layers together before they are impregnated with the matrix material [6]. Z-pinning is more effective, where small rods or fibres are inserted through the thickness before curing to resist delamination and bridge cracks [7]. It does introduce defects, so depending on the design requirements, this trade-off may be acceptable.

Replacing manual layup with automated tape placement (ATP) or automated fibre placement (AFP) allows fibre paths to be varied around geometric discontinuities such as holes. Instead of straight fibres being cut by the drilled hole, fibres can be steered around the opening, reducing the number of cut load paths and lowering peak stresses [8]. These variable-stiffness laminates redistribute load more efficiently over a larger area. AFP-based tow steering has been shown in a NASA study to reduce Tsai–Wu failure stresses significantly [9]. However, there are several challenges with ATP and AFP, such as high tooling costs and temperature control during layup to minimise void content. Post-processing is usually required due to the heat applied locally during layup, which causes inhomogeneous crystallinity and stress concentrations between the layers.

Alternative fibre architectures to 2D unidirectional laminates can also provide improved performance. Woven fabrics interlace fibres in warp and weft directions, enabling loads to be shared between fibre bundles. This reduces the impact of matrix cracking and shear-driven failure, while the interlaced structure provides improved delamination resistance and better tolerance to drilling damage. Although fibre crimp in woven fabrics slightly reduces axial stiffness, woven composites exhibit higher bearing strength and multidirectional load-carrying capacity. Weaves include plain weave, twill, satin, and basket weaves [10].

Finally, 3D woven composites offer substantial improvement due to the presence of through-thickness reinforcement. Interlocking tows in the x, y, and z directions overcome many interlaminar weaknesses of 2D laminates [11] [12]. Delamination resistance is increased significantly, and cracks that would normally propagate along ply interfaces are resisted by the z-direction fibres. Again, 3D woven composites may reduce stiffness, but they significantly improve damage tolerance and load capacity, suitable for components containing holes.

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